

THE LEAN HANDBOOK



**A GUIDE TO THE BRONZE CERTIFICATION
BODY OF KNOWLEDGE**

**ANTHONY MANOS
AND
CHAD VINCENT, EDITORS**

THE LEAN HANDBOOK

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A Guide to the Bronze Certification
Body of Knowledge

*Anthony Manos and
Chad Vincent, editors*

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The Global Voice of Quality™

To Jennifer, who puts up with all my antics. To Judy, who taught me her version of continuous improvement—"Do not rest on your laurels." And to George, for supporting me and for helping me reach a higher level of accomplishment.

—Tony Manos

To my wife, Holly, and my kids, Austin and Miranda, who gave up their time with a husband and father to make this book possible, and whose continued support provides me strength to pursue my dreams.

—Chad Vincent

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Foreword

Robert D. Miller

Executive Director

The Shingo Prize for Operational Excellence

Jon M. Huntsman School of Business

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The *Lean Handbook* is a terrific compilation of sections written by practitioners who bring their individual and unique experience and expertise to this body of knowledge. I appreciate the connections made with many of the dimensions and principles identified in the Shingo model. We must each be on a journey of continuous improvement, which means we must be constantly looking for new perspectives and approaches to bring about personal and organizational improvement.

Anthony Manos is correct when he says that the most difficult and yet important work we will do with lean is to change the culture of the organization. For lean to be successful, the many tools outlined in this handbook must be supplemented with an equally devoted effort to influence the mind-sets and behaviors of people in the organization. At the Shingo Prize we have learned that the best way to do this is to, as Stephen Covey would say, “begin with the end in mind.” In other words, while you are learning to use these great tools, you must also keep the deeper meaning, or the “why” behind the tools, very clear. *The Lean Handbook* can help each of us act our way into a new way of thinking, and then think our way into a new way of acting. By putting the wisdom of this handbook to work every day by every person and never taking your eyes off the prize—a new culture that is deeply embedded in the principles of lean—we will greatly increase the odds of a sustainable business transformation.

As you put this work into practice, you will recognize the shifting roles of leaders and managers in your organization. It is not enough for leaders to just keep doing what they have always done, nor is it enough for them to merely support the work of others. Rather, leaders must lead the cultural transformation and build the principles behind all of these great tools into the mind-sets of their associates. Similarly, managers have to do more than participate in kaizen teams. The emerging role of managers is to focus on designing, aligning, and improving the systems of the business so that they drive ideal behaviors that cause people to change their thinking of what excellence really looks like.

Using *The Lean Handbook* as a roadmap will no doubt be a powerful tool in helping you avoid many of the mistakes made by others over the years. I invite you to visit <http://www.shingoprize.org> to see how the key points illustrated here support the Shingo model for operational excellence. My thanks to all of the contributing authors!

Preface

WELCOME TO *THE LEAN HANDBOOK*

What a remarkable journey this has been. Working on this book has been a terrific experience. We have had the great pleasure of working with a number of wonderful and giving individuals. Lean practitioners are truly an amazing and unique family. The energy and willingness of the individuals who helped create this book are evidence of the great profession and network of people of which we take part. So many different points of view and applications of knowledge made for great discussions, contemplation, and collaboration. With so much knowledge and understanding, it was difficult to find a point at which to stop talking and start putting these discussions on paper. In the end, we believe this book embodies the Lean Body of Knowledge (BOK) in a way that is much like the lean journey—ever evolving and always adaptable.

Lean has been a culmination of multiple individuals, philosophies, systems, tools, and applications throughout history. The challenge has been that all these different contributions are found in different places, called different things, and applied in different manners—making it difficult for the lean practitioner to gain an understanding of lean at a level of its full body of knowledge without great effort, research, experience, and networking. While this book is not a substitute for the effort, research, experience, and networking every seasoned lean practitioner goes through, we hope it provides a sound starting point for those just beginning or expanding their knowledge of lean.

NOT AN EXAM PREPARATION MANUAL

First and foremost, this book is not a Lean Certification exam preparation manual. The Lean Bronze Certification exam questions are based on material from the five recommended reading books (see Appendix B, “Recommended Reading List for Lean Certification Exam Preparation”):

- *Learning to See*
- *Lean Thinking*
- *Gemba Kaizen*
- *Lean Production Simplified*
- *Lean Hospitals*

Make no mistake—this book takes nothing away from the great lean works that have preceded it. As a matter of fact, we believe that this book complements and pays tribute to those works as being pieces of the larger Lean BOK. But that is exactly what they are—pieces. Our intent was to put these pieces together in a manner to provide a higher-level overview of the Lean BOK. We realized early on in the project that this task was not something we could do alone.

This handbook's intention is to gather information related to the Lean BOK (see Appendix A, "Lean Certification Body of Knowledge") into one source. This book will enhance your understanding of the BOK as a whole and give you a more holistic look at lean. As great as the five recommended reading books are, they were not written with the intent of covering all aspects of the Lean BOK individually. Additionally, this book does not rehash the content of the five recommended reading books. What we have done is put together a book whose sole purpose is to embody the entire Lean BOK, section by section. This book is, by design, written at the Bronze Level for certification knowledge. This means that the weightings used in the Lean BOK for the Bronze Certification were considered for the depth and breadth of material considered for each rubric. Therefore, it is by no means all inclusive of every principle, system, and tool at every level of application related to lean.

By addressing the Lean BOK at the Bronze Level, this book provides a basic understanding of the lean principles, systems, and tools at a tactical level to drive improvements with measureable results. The intent is to revise the book over time to encompass the topics of the Silver Level (an integrated application on value stream transformations for lean leaders) and the Gold Level (for strategic application of lean across the entire enterprise, with emphasis on assets, systems, processes, and people). Therefore, this book, much like a lean journey in an organization, will be adapted as the Lean BOK evolves and more knowledge is integrated.

Given that the intent of this book is not to rehash the certification reference books, we hope that this book serves as a good starting point for those practitioners who want a holistic view of the Lean BOK, with links to many other lean references for greater detail and understanding. While there are many references, we tried to stay true to the terminology and applications discussed in the core books of the certification reading list.

One of the difficulties we faced in creating a book of this magnitude was how to structure it. While we could have structured it alphabetically by topic, organized it by case studies or by organizations, or arranged it by some other method, we wanted to stay true to the Lean BOK structure. While this structure does not allow for a nice flow from one topic to another for easier reading, the writing conforms to the Lean BOK and the Shingo Prize model. We thought that this would provide a traceable reference for those individuals and organizations utilizing those structures for the pursuit of operational excellence.

When lean is applied in an organization, the knowledge of the processes and the generation of ideas do not come from the organization's designated lean experts. They come from those who perform the work on a daily basis. We took the same approach with this book. It would have been easy for us to read all the books and then pull information from those books to create another book. But then it would have been just that—another book. We needed to take a lean approach with this book. So, just as you would create a team of individuals who perform the

work in a kaizen event, we assembled a team of individuals who perform the work and who apply lean in their organizations every day.

MANY VOICES AND MANY STYLES

We were lucky to have some of the best minds in lean contribute to this endeavor (see “Contributing Authors and Editors”). There are many voices, many contributors, many styles of writing, and more than one point of view. The contributing authors come from many different backgrounds. Such different life experiences weave a wonderful lean tapestry. This book is not just lean for manufacturing—or lean for service or lean for healthcare. The examples given in this book can fit any type of organization. We hope you find these different points of view helpful while finding your voice in lean.

It has been a pleasure to not only be authors and share our knowledge of lean but also be editors and work closely with others like us. The great thing about working with all these individuals was learning how they apply the same things we apply, but maybe just a little differently. These differences provided us a different perspective on our version of lean and were wonderful opportunities to expand our personal lean knowledge base.

Coming together is a beginning. Keeping together is progress. Working together is success.

—Henry Ford

We wish you the best of luck and fair weather on your lean journey!

Acknowledgments

It would be nearly impossible to mention everyone who had an influence on the creation of this lean handbook, but we would like to make a few special mentions.

First and foremost we would like to thank all the contributing authors, who worked tirelessly to help construct and shape this handbook. Their willingness to share their knowledge and experience was exceptional. To learn more about these extraordinary lean thinkers, see “Contributing Authors.” A special acknowledgment goes to all the people and organizations that the contributing authors and editors have worked with over the years to help develop and deepen our understanding of lean and influence us as we continue to learn more.

This handbook would not have been possible without the support of the Lean Enterprise Division (LED) of the American Society for Quality (ASQ) and the LED Leadership Team—Kiami Rogers (chair), Frank Murdock (chair-elect), and Tammy Miller (secretary). We would also like to thank George Alukal, founding member of the LED and the driving force behind lean’s becoming an integral part of ASQ and a resource for its members.

The Lean Certification is supported by the four alliance partners: the Society of Manufacturing Engineers (<http://www.sme.org>), the Association for Manufacturing Excellence (<http://www.ame.org>), the Shingo Prize (<http://www.shingo.prize.org>), and of course ASQ (<http://www.asq.org>).

We would like to thank Kris Nasiatka from the Society of Manufacturing Engineers (SME) for all her efforts in creating the Lean Certification and for her continued support of the partner organizations and this lean handbook. Also from SME, Kelly Lacroix leads the Lean Certification Oversight and Appeals Committee, which continually monitors and improves the certification process.

If it wasn’t for our friends Matt Meinholz and Paul O’Mara at ASQ Quality Press, this book never would have been completed. We appreciate their patience and advice while working on this endeavor.

Last, but not least, we would like to give our utmost gratitude to Robert Damelio. As a member volunteer, Robert not only was the driving force behind ASQ’s LED adopting the Lean BOK, but he also guided the certification initiative at ASQ. Without his tireless efforts, ASQ would not have been a partner member of the Lean Certification.

A Brief History of the Lean Certification Body of Knowledge

Kris Nasiatka, SME

HOW IT STARTED

The publication of *The Machine that Changed the World*, in 1990, brought the concept of lean to the masses. It also provided opportunities for many different flavors of lean to be born. With that came a myriad of education, training, and consulting practices, each bringing its own version of lean to the market.

In 2001, members of the Society of Manufacturing Engineers (SME) and the Association for Manufacturing Excellence (AME), and constituents of the Shingo Prize came together and determined that some type of validation for professional practice of lean was necessary. There was a need to align practitioners with a common foundation (fundamentals) of lean practice and, more importantly, provide a roadmap to support workforce development and training efforts. After a few stops and starts, development was under way in earnest in 2004.

The parties involved agreed that the Shingo Prize model should serve as the basis for the program's Body of Knowledge (BOK). The initial BOK went deeply into evaluating lean practitioners' job tasks. The model was modified to make it applicable to people and workforce development efforts versus being a corporate lean transformation model. The proposed Lean BOK was validated in a study in early 2005, thus launching version 2.0.

KAIZEN BLITZ WEEK

Shortly after version 2.0 was launched, a core development committee was established. With the preliminary validation of the BOK in place, a straw man for program components was outlined. SME hosted a "blitz week" to expedite development of the program. Nearly 60 lean practitioners, representing a breadth of manufacturing industries, consulting practices, and academia, gathered for a week to develop the components of this new certification program. During the blitz, development teams were established to focus on exams, portfolios, and the mentoring requirements for this new program. Throughout the entire development process, nearly 200 lean practitioners were involved in the creation of this program not including the couple of thousands that participated in the study.

The first Bronze exam was offered to the public in March 2006. The Silver exam was available in June 2006, and the Gold exam was launched in December of that year. The formal Lean Certification Oversight and Appeals Committee was established and met for the first time in March 2006.

UPDATED VERSION OF THE BOK

With the program “live” for a year, a second BOK validation study was conducted. It was desired to structure the certification BOK similar to the Shingo Prize model. The Shingo Prize was restructuring its model, which created an opportunity for the certification program to validate proposed changes to the BOK received by certification candidates as well as the oversight body.

The validation study was completed in 2007. The Lean Certification Oversight and Appeals Committee used the findings to restructure the BOK and launched version 3.0 in 2008 (see Appendix A). Although the topics did not change, the BOK was reorganized for better flow and improved alignment of topics within each category. Each BOK validation study is structured in a way that derives a weighting factor for each section of the BOK. The weighting factors identify the percentage of exam coverage for each BOK area and help the committee identify priority areas for building the exam bank.

THE FUTURE OF THE BOK

Validation studies will be conducted every five to seven years. The studies are intended to verify that current topics are still relevant in contemporary practice of lean and to identify any new or emerging topics that should be added to the BOK.

Special Dedication

Kiami Rogers
Chairperson, ASQ Lean Enterprise Division

This handbook is dedicated in loving memory to Wayne Paupst (1957–2010), past chairman of the Lean Enterprise Division (LED). In August 2010, Wayne lost a long battle with cancer. Wayne was a quality professional's "quality professional." He never complained about his condition. In fact, many of us on the Leadership Team, which worked closely with Wayne, were not even aware of the severity of his condition. Wayne possessed a wonderful sense of humor and had a kind word for everyone. He was always ready with a joke, and always ready to laugh at jokes offered by others. His leadership, instruction, kindness, and humor will be missed by family and friends as well as organizations such as ASQ.

I first met Wayne at a meeting of the founders of the LED prior to the LED becoming a forum and subsequently a division. Wayne had been a member of ASQ since 1988. He had more than 25 years in the quality profession, holding positions such as quality engineer, quality systems coordinator, inspector, and quality assurance manager. He had been instructing certification courses for the Lehigh Valley Section of ASQ since 1996 and also had provided instructional courses for many of the top companies in the Lehigh Valley as well. Wayne also held several ASQ certifications: Six Sigma Black Belt, Quality Engineer, Quality Manager, Quality Auditor, Quality Technician, Quality Inspector, Process Analyst, and Quality Improvement Associate. It was during Wayne's tenure as LED division chair that ASQ introduced the Lean Certification, in partnership with SME, AME, and the Shingo Prize organizations. I have no doubt that Wayne would have pursued obtaining the ASQ Lean Certification himself, as well as teaching and mentoring other lean professionals pursuing this certification. It is with great honor that we dedicate this handbook to Wayne.

Prologue

Kiami Rogers

Chairperson, ASQ Lean Enterprise Division

The purpose of this handbook is to provide a reference guide for lean principles and methods. This handbook on its own is not intended to prepare one for the ASQ Lean Certification (in partnership with SME, AME, and the Shingo Prize organizations). The user of this handbook is a lean professional who has some knowledge of and experience with lean principles and methods. Material from several lean practitioners with differing levels of disclosure of their experiences has been gathered to create this handbook and has been edited to be presented in a consistent and unified format.

With 6000 members worldwide, the ASQ Lean Enterprise Division (LED) is a global network of professionals helping individuals and organizations apply proven and leading-edge lean principles and practices to achieve dramatic results for personal and organizational success. Whether or not you are a member of the ASQ LED, we hope you find this handbook a useful guide in your lean journey.

Contributing Authors and Editors

This handbook was a large collaborative effort, and we would especially like to thank all the contributing authors, who shared their time and wisdom to help make this endeavor possible.

CONTRIBUTING AUTHORS

Christopher Abrey is a program manager with Northstar Aerospace in Bedford Park, Illinois. He earned a bachelor of engineering in manufacturing systems engineering from Coventry University, UK. He is a senior member of the Society of Manufacturing Engineers (SME) and a member of the Association of Manufacturing Excellence (AME). Christopher has more than 20 years of lean experience in manufacturing and recently completed the Lean Bronze Certification. He can be reached at cabrey@nsaero.com.

Section

2.2.5. Total Productive Maintenance (including predictive)

Andy Carlino is cofounder and partner of the Lean Learning Center. In addition to over 20 years' experience in real-world senior management, including president and COO, he has for the past 20 years been providing consulting, training, and coaching services from the boardroom to the shop floor to organizations across the globe. He has published numerous articles and is coauthor of *The Hitchhiker's Guide to Lean: Lessons from the Road*. Andy is a frequent speaker for a variety of professional organizations and corporate conferences, including guest speaker for Clemson and Harvard University. He holds a BS in engineering and a BA in psychology and is a member of SME, SAE, AME, ASQ, and numerous other industry and professional associations.

Sections

1.2.1. Planning & Deployment

1.2.2. Create a Sense of Urgency

1.2.3. Modeling the Lean Principles, Values, Philosophies

1.2.7. Motivation, Empowerment & Involvement

Adil Dalal is CEO of Pinnacle Process Solutions, Intl.; a keynote speaker; thought leader in lean, project management, and leadership; and author of *The 12 Pillars of Project Excellence: A Lean Approach to Improving Project Results*. He has earned MS degrees in engineering management and mechanical engineering. He is a Certified Lean Bronze Professional, Certified Quality Engineer (CQE), Project Management Professional (PMP), and Board Certified Executive Coach. He is the chair of ASQ's Human Development & Leadership Division, and certification chair of the Lean Enterprise Division. He served on the Lean Certification Oversight and Appeals Committee. He can be reached at adil@pinnacleprocess.com.

Sections

- 1. Cultural Enablers
 - 1.1. Principles of Cultural Enablers
 - 1.1.1. Respect for the Individual
 - 4.3.5. Competitive Impact

Grace Duffy is president of Management and Performance Systems. She holds an MBA from Georgia State University and has coauthored numerous books: *The Quality Improvement Handbook*, *Executive Guide to Improvement and Change*, *Executive Focus: Your Life and Career*, *The Public Health Quality Improvement Handbook*, *QFD and Lean Six Sigma for Public Health*, *Modular Kaizen: Dealing with Disruption*, and *Tools and Applications for Starting and Sustaining Healthy Teams*. She is an ASQ Certified Manager of Quality/Organizational Excellence (CMQ/OE), Improvement Associate, Auditor, Lean Six Sigma Master Black Belt (LSS MBB), and ASQ Fellow.

Section

- 2.2.8. Strategic Business Assessment

David S. Foxx is a senior manager and Champion of the Continuous Process Improvement Community of Practice at Deloitte Consulting. He earned an MBA at the University of Phoenix and a bachelor of science in industrial engineering at the University of Texas at El Paso. David is a leader in enterprise transformation and specializes in innovation, product, process, and service design. David presented "Total Quality Design—An Imperative for Survival" at the Annual Quality Congress. He is a senior member of ASQ and a member of the ASQ Quality Press Standing Review Board, the Institute of Industrial Engineers (IIE), and the International Society of Six Sigma Professionals (ISSSP). His certifications include ASQ Certified Six Sigma Black Belt (CSSBB), LSS MBB, Lean Master, and Design and Innovation Master Black Belt. He can be reached at sixsigmalean@aol.com.

Sections

- 2.1.2.2. 7 Wastes (Muda), Fluctuation (Mura), and Overburden (Muri)
- 2.3.5. Product and Service Design
 - 2.3.5.1. Concurrent Engineering

- 2.3.5.2. Quality Function Deployment
- 2.3.5.3. Product or Process Benchmarking
- 2.3.5.4. Design for Product Life Cycle (DFx)—Cradle to Cradle
- 2.3.5.5. Variety Reduction—Product and Component
- 2.3.5.6. Design for Manufacturability
- 4.3.2. Delivery
 - 4.3.2.1. Takt Time
 - 4.3.2.2. Cycle Time
 - 4.3.2.3. Lead Time

Dr. Gwendolyn Galsworth is president and founder of Visual Thinking, a training, research, and consulting firm, and the VTI/Visual-Lean Institute, which offers licensing and train-the-trainer in nine core visual workplace courses. Over some 30 years of hands-on implementations, Gwendolyn has nearly single-handedly created the models, concepts, and methods of workplace visuality that define visual's distinct and powerful contribution to enterprise excellence—and to sustainable cultural and bottom-line results. Gwendolyn serves as a Shingo Prize examiner. She is the author of *Visual Systems: Harnessing the Power of the Visual Workplace*; *Smart, Simple Design: Using Variety Effectiveness to Reduce Total Cost and Maximize Customer Selection*; two Shingo award-winning books, *Visual Workplace*, *Visual Thinking: Creating Enterprise Excellence through the Technologies of the Visual Workplace* and *Work That Makes Sense*; and many DVDs. She can be reached at gwendolyn@visualworkplace.com.

Section

2.2.1. Visual Workplace

Bruce Hamilton is president of the Greater Boston Manufacturing Partnership, headquartered at the University of Massachusetts Boston campus. He attended Bowdoin College and earned a BA from the University of Arizona. Bruce is the creator of *Toast Kaizen* and other lean training videos, and also coauthor with Pat Wardwell of the *e² Continuous Improvement System Manual*. He posts weekly to his blog: <http://www.oldleandude.org>. Both a recipient of the Shingo Prize and an inductee into the Shingo Prize Academy, Bruce is also a Shingo Prize examiner and a member of the Shingo Prize Board of Governors. He can be reached at pokayoke@comcast.net.

Sections (with Pat Wardwell)

- 2.1.2. Identification & Elimination of Barriers to Flow
 - 2.1.2.1. Flow & the Economies of Flow
 - 2.1.2.3. Connect & Align Value Added Work Fragments
 - 2.1.2.4. Organize around Flow

- 2.1.2.5. Make End-to-End Flow Visible
- 2.1.2.6. Manage the Flow Visually
- 2.3.7.1. Mistake and Error Proofing (Poka Yoke)
- 2.3.7.4. Right Sized Equipment
- 2.3.7.5. Cellular Flow

John Kendrick is a principal with Fujitsu in Sunnyvale, California, and has more than 15 years of lean experience in manufacturing, finance, telecommunications, and healthcare. John holds a master of engineering degree in simulation and modeling from Arizona State University, a master of applied statistics from Penn State, and a master of business administration in finance from the University of Pittsburgh. He is a Certified Six Sigma Master Black Belt (CSS MBB) and a senior member of ASQ. He is also a CSSBB, Certified Reliability Engineer (CRE), Certified Software Quality Engineer (CSQE), and CMQ/OE and holds two Lean Certifications.

Sections

- 2.3.1. Work Flow Analysis
 - 2.3.1.1. Flowcharting
 - 2.3.1.2. Flow Analysis Charts
 - 2.3.1.3. Value Stream Mapping
 - 2.3.1.4. Takt Time Analysis
- 4.2.1. Measurement
 - 4.2.1.1. Understand Interdependencies between Measures and Measurement Categories
 - 4.2.1.2. Align Internal Measures with What Matters to Customers
 - 4.2.1.3. Measure the Results from the “Whole” System
 - 4.2.1.4. Measure Flow and Waste
- 4.2.3. Analysis—Understand What Moves the Dial on Measures
- 4.2.4. Reporting
 - 4.2.4.1. Visible Feedback Real-Time

Matthew Maio is a quality manager at Raytheon IDS in White Sands, New Mexico. He earned bachelor’s degrees in business and computer science from the College of Santa Fe, New Mexico. He is the author of *Quality Improvement Made Simple . . . and Fast!* and coauthor of *The Six Sigma Green Belt Handbook* (both from ASQ Quality Press). Matthew is a member of the International Test and Evaluation Association and the Directed Energy Professionals Society. He is a senior member of ASQ and past regional director (board member) and section chair of ASQ. He holds ASQ certification as a CQM/OE, CSSBB, Certified Six Sigma Green Belt (CSSGB),

CSQE, and Certified Quality Auditor (CQA) and Defense Acquisition University Lean Six Sigma Yellow Belt (LSSYB) and Process Quality Management (PQM) certification/recognition. He can be reached at Matt_Maio@comcast.net.

Sections

- 1.2.4. Message Deployment—Establishing Vision and Direction
- 1.2.5. Integrating Learning and Coaching
- 1.2.6. People Development—Education, Training & Coaching
- 1.2.8. Environmental Systems
- 1.2.9. Safety Systems
- 2.1.5.1. Quality at the Source
- 2.1.5.4. Multi-Process Handling
- 2.2.6. Standard Work
- 2.2.7. Built-in Feedback
- 2.2.15. Knowledge Transfer
- 2.3.7.3. One Piece Flow
- 2.3.7.4. Right Sized Equipment
- 2.3.7.6. Sensible Automation
- 2.3.7.8. Source Inspection

David Mann is principal at David Mann Lean Consulting. He is the Shingo Prize-winning author of *Creating a Lean Culture: Tools to Sustain Lean Conversions* (Productivity Press). David serves on the editorial board of AME's publication, *Target*, on the management science faculty at the Fischer College of Business at Ohio State University, and as a Shingo Prize examiner. He earned his PhD in psychology from the University of Michigan. He can be reached at dmann@dmannlean.com.

Sections

- 2.1.1. Process Focus
- 3.1.1.3. Closed-Loop Thinking to Assure Effective Feedback of Organizational Learning

Anthony Manos is a catalyst with Profero and lean champion at 5S Supply in Frankfort, Illinois. He earned an MBA in entrepreneurial studies from the University of Illinois at Chicago. He is the coauthor of *Lean Kaizen: A Practical Approach to Process Improvement* (ASQ Quality Press), a senior member of ASQ, senior member of SME, cofounder and past chair of ASQ's Lean Enterprise Division, and a member of the Lean Certification Oversight and Appeals Committee. He is Lean Bronze Certified. He can be reached at anthony.manos@proferoinc.com.

Sections

- 1.1.2. Humility
- 1.3.5. Coaching & Mentoring
- 2.2.1.1. 5S Standards and Discipline
- 3.2.2. Policy Deployment/Strategy Deployment
- 4.3.5.1. Customer Satisfaction

Brian H. Maskell is president of BMA in Cherry Hill, New Jersey, and St. Albans, Hertfordshire, England. He has an engineering degree from the University of Sussex, England. Brian is certified with the Chartered Institute of Management Accountants (CIMA) in London, the American Institute of Certified Public Accountants (AICPA), and the Institute of Management Accountants (IMA). He is a Fellow of the American Production and Inventory Control Society (APICS). Brian is the author of eight books, including *Making the Numbers Count: The Accountant as Change Agent on the World Class Team* (second edition) and *Practical Lean Accounting: A Proven System for Measuring and Managing the Lean Enterprise* (second edition). He can be reached at bmaskell@maskell.com.

Section

- 4.2.1.5. Lean Accounting

Timothy F. McMahon is the founder of and a contributor to *A Lean Journey Blog* (<http://aleanjourney.com>). He has a BS in chemical engineering from the University of Massachusetts and holds a Lean Certification and Six Sigma Black Belt from Central Connecticut State University. Tim is a member of the regional board of directors for AME and currently serves as the VP of program for northeast region. He can be reached at Tim@aleanjourney.com.

Sections

- 1.3.1. Cross Training
- 1.3.3. Instructional Goals
- 1.3.6. Leadership Development
- 1.3.7. Teamwork
- 1.3.8. Information Sharing (Yokoten)
- 1.3.9. Suggestion Systems
- 2.2.14. Pull System
- 2.3.6. Organizing for Improvement
 - 2.3.6.1. Kaizen Blitz Events
- 2.3.7. Countermeasure Activities
 - 2.3.7.1. Mistake and Error Proofing (Poka Yoke)

2.3.7.2. Quick Changeover/Setup Reduction (SMED)

2.3.7.3. One Piece Flow

2.3.7.4. Right Sized Equipment

2.3.7.5. Cellular Flow

2.3.7.7. Material Signals (Kanban)

Dr. Mark W. Morgan is the associate vice president for institutional effectiveness and research at Seminole State College of Florida. Mark earned his doctorate in educational leadership from the University of Florida and is a three-time examiner for the Malcolm Baldrige National Quality Award. Mark was an improvement consultant for Fortune 500 companies for more than 10 years and is the author of three books on measurement and performance improvement, including his latest, *The Path to Profitable Measures: 10 Steps to Feedback That Fuels Performance* (ASQ Quality Press).

Section

4.1. Principles of Business Results

Frank Murdock is senior process engineer at Plymouth Tube Company in West Monroe, Louisiana. He earned a BS in engineering science at Purdue University, an MS in applied mathematics at the University of Michigan, and an MS in industrial engineering at Wayne State University. Frank spent 28 years with the Ford Motor Company, 8 years as an independent consultant, and 6 years as an adjunct professor at Lawrence Technological University. A senior member of ASQ and an ASQ Certified Six Sigma Black Belt, Frank is chair-elect for the Lean Enterprise Division as well as chair of the ASQ Voice of the Customer Committee. He can be reached at fmurdock@plymouth.com.

Section

4.2.1.6. Voice of the Customer

Mike Osterling has been a lean management practitioner and leader since the mid-1990s and is the lead consultant at Osterling Consulting, a San Diego-based firm. He earned an MBA in international business at San Diego State University and a BS in production and operations management. Mike coauthored *The Kaizen Event Planner: Achieving Rapid Improvement in Office, Service, and Technical Environments*. He is a certified trainer for the implementation of lean manufacturing (University of Kentucky) and a Lean Six Sigma Black Belt (University of California San Diego). He is certified in production and inventory management (APICS). He can be reached at mike@mosterling.com.

Sections

2.1.5. Jidoka

2.1.7. Seek Perfection

Robert (Bob) Petruska works as a performance improvement consultant in Charlotte, North Carolina. He has a master of science degree in manufacturing systems from Southern Illinois University. Bob is a senior member of ASQ and a CSSBB. Bob is authoring a book titled *Gemba Walks for Service Excellence: The Step-by-Step Guide for Identifying Service Delighters*, to be published in 2012 by Productivity Press.

Sections

- 3.2.1. Enterprise Thinking
 - 3.2.1.1. Organize around Flow
 - 3.2.1.2. Integrated Business System and Improvement System
 - 3.2.1.3. Reconcile Reporting System
 - 3.2.1.4. Information Management

Govind Ramu is a senior manager for global quality systems for SunPower Corporation. Prior to this role, he was Six Sigma Master Black Belt for JDS Uniphase Corporation. Govind is a professional engineer (mechanical) from Ontario, Canada, and an ASQ Fellow. He holds six ASQ certifications. Govind has had articles published in *Quality Progress* and in the Six Sigma forum. He coauthored ASQ's *The Certified Six Sigma Green Belt Handbook*, for which he received the ASQ Golden Quill Award in 2008. Additionally, Govind was a 2006 and 2011 examiner for the California Awards (CAPE) and a 2010 examiner for the Malcolm Baldrige Award. He can be reached at ramu.govind@gmail.com.

Sections

- 2.1.5.1. Quality at the Source
- 2.1.7.1. Incremental Continuous Improvement (Kaizen)
- 2.2.9. Continuous Improvement Process Methodology
 - 2.2.9.1. PDCA
 - 2.2.9.2. DMAIC
 - 2.2.9.3. Problem Solving Storyboards
- 2.2.10. Quality Systems
- 2.2.11. Corrective Action System
- 2.3.3.3. Failure Mode and Effects Analysis
- 2.3.4. Presenting Variation Data
 - 2.3.4.1. Statistical Process Control Charts
 - 2.3.4.2. Scatter and Concentration Diagrams
- 2.3.5.2. Quality Function Deployment
- 2.3.5.4. Design for Product Life Cycle (DFx)—Cradle to Cradle
- 2.3.5.6. Design for Manufacturability

Rama Shankar is the managing partner at Delta Management Associates in Glenview, Illinois. She has a master's degree in engineering management from Northwestern University and a master's degree in materials management from Indian Institute of Materials Management, India. Rama is the author of *Process Improvement Using Six Sigma: A DMAIC Guide* (ASQ Quality Press). She is a senior member of ASQ and a past chair, section director, and training institute director of the ASQ Chicago Section. Rama is a past Malcolm Baldrige Award examiner, an ASQ CSSBB, and a CQA. She is also a certified trainer by NIST for lean.

Sections

2.1.5. Jidoka

2.1.5.1. Quality at the Source

2.1.5.2. No Defects Passed Forward

2.1.5.3. Separate Man from Machine

2.1.5.4. Multi-Process Handling

2.1.5.5. Self Detection of Errors to Prevent Defects

2.1.5.6. Stop and Fix

With Tim McMahon

1.3.2. Skills Assessment

1.3.4. On-the-Job Training

1.3.6. Leadership Development

1.3.7. Teamwork

With Govind Ramu

2.2.9.1. PDCA

2.2.9.2. DMAIC

2.2.9.3. Problem Solving Storyboards

2.2.11. Corrective Action System

2.2.11.1. Root Cause Analysis

2.2.12. Project Management

2.3.2. Data Collection and Presentation

2.3.2.1. Histograms

2.3.2.2. Pareto Charts

2.3.2.3. Check Sheets

2.3.3. Identify Root Cause

2.3.3.1. Cause & Effect Diagrams (Fishbone)

2.3.3.2. 5-Whys

2.3.3.3. Failure Mode and Effects Analysis

2.3.4. Presenting Variation Data

2.3.4.1. Statistical Process Control Charts

2.3.4.2. Scatter and Concentration Diagrams

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Sections

2.2.4. 3P Production Process Preparation

2.2.13. Process Design

3.1.1. Systemic Thinking

3.1.2. Constancy of Purpose

3.1.3. Social Responsibility

4.2.2. Goal and Objective Setting

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Sections

2.1. Principles of Continuous Process Improvement

2.1.2.2. 7 Wastes (Muda), Fluctuation (Mura), and Overburden (Muri) (with David Foxx)

2.1.5.1. Quality at the Source (with Govind Ramu and Mark Paulson)

2.1.5.3. Separate Man from Machine

2.1.6. Integrate Improvement with Work

2.1.7. Seek Perfection

2.2. Continuous Process Improvement Systems

2.2.2. Lot Size Reduction

2.2.3. Load Leveling

- 2.2.4. 3P Production Process Preparation
- 2.2.10. Quality Systems
 - 2.2.10.1. ISO and Other Standards
- 2.2.11. Corrective Action System
- 2.3. Continuous Process Improvement Techniques & Practices
 - 2.3.1. Work Flow Analysis
 - 2.3.1.1. Flowcharting (with John Kendrick)
 - 2.3.1.2. Flow Analysis Charts (with John Kendrick)
 - 2.3.1.3. Value Stream Mapping
 - 2.3.1.4. Takt Time Analysis (with Matt Maio)
 - 2.3.8. Supplier Processes External
 - 2.3.8.1. Supplier Managed Inventory
 - 2.3.8.2. Cross-Docking
 - 2.3.8.3. Supplier Assessment and Feedback
 - 2.3.8.4. Supplier Development
 - 2.3.8.5. Supplier Benchmarking
 - 2.3.8.6. Logistics
 - 2.3.9. Supply Processes Internal
 - 2.3.9.1. Material Handling
 - 2.3.9.2. Warehousing
 - 2.3.9.3. Planning and Scheduling
- 4.3. Key Lean Related Measures
 - 4.3.1. Quality
 - 4.3.1.1. Rework
 - 4.3.1.2. First Pass Yield
 - 4.3.3. Cost
 - 4.3.3.1. Inventory Turns
 - 4.3.3.2. Queue Time
 - 4.3.3.3. Wait Time (delays)
 - 4.3.3.4. Overall Equipment Effectiveness (OEE)
 - 4.3.3.5. Changeover Time
 - 4.3.4. Financial Impact
 - 4.3.4.1. Cash Flow

Pat Wardwell is the chief operating officer at Greater Boston Manufacturing Partnership in Boston, Massachusetts. She holds a BA from the University of Maine and an MBA from Bentley College. Pat is the coauthor of *e² Continuous Improvement System*. Her accomplishments include SME Lean Gold Certified, past chair of the Lean Certification Oversight and Appeals Committee, Shingo Prize recipient and examiner, AME Manufacturing Excellence Awards committee member and examiner, AME Northeast Board of Directors, and member of SME Boston Chapter Leadership Committee. She can be reached at pwdardwell@gbmp.org.

Sections (with Bruce Hamilton)

2.1.2. Identification & Elimination of Barriers to Flow

2.1.2.1. Flow & the Economies of Flow

2.1.2.3. Connect & Align Value Added Work Fragments

2.1.2.4. Organize around Flow

2.1.2.5. Make End-to-End Flow Visible

2.1.2.6. Manage the Flow Visually

2.3.7.1. Mistake and Error Proofing (Poka Yoke)

2.3.7.4. Right Sized Equipment

2.3.7.5. Cellular Flow

Jerry M. Wright, P.E., is the senior vice president of lean and enterprise excellence for DJO Global in Vista, California. He earned an MBA in 2002 from the University of Phoenix and is a registered professional engineer in the state of California. He is the annual international conference chair for AME for Chicago in 2012 and also a west region director for AME. He is also the chair of the So Cal Lean Network, an affiliation of more than 75 companies focused on lean and sharing in Southern California, as well as a Shingo Prize examiner and a previous Baldrige Award examiner. He can be reached at jerry.wright@djoglobal.com.

Sections

3.3.1. A3

3.3.2. Catchball

3.3.3. Redeployment of Resources

EDITORS

Anthony Manos is a catalyst with Profero, where he provides professional consulting services, implementation, coaching, and training for a wide variety of organizations (large and small, private and public) in many industries focusing on lean enterprise and lean healthcare. Tony has extensive knowledge of lean and quality in a wide range of work environments. He is trained and certified by the National

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Relying on his diverse knowledge of business, manufacturing techniques, and applications, Tony has assisted numerous clients in implementing a lean enterprise and lean healthcare. Over the past 10 years, he has helped over 150 companies in several aspects of lean implementation, including team building, standardized work, 5S workplace organization and visual workplace, quick changeover, plant layout, cellular, kanban, total productive maintenance, kaizen events, and hoshin planning. Tony is an internationally recognized speaker and expert on lean and presents at several conferences a year. As an ASQ faculty member, he teaches a two-day course in lean enterprise and a one-day course on kaizen. Tony is the past chair of the Lean Enterprise Division of ASQ. He is a senior member of SME and a member of AME. Tony is coauthor of the book *Lean Kaizen: A Simplified Approach to Process Improvement* and author of many articles on lean and its allied subjects. He serves as an ASQ representative to the Lean Certification Oversight and Appeals Committee. Tony is Lean Bronze Certified and was part of the original team to develop the Lean Certification.

Tony served in the US Navy nuclear propulsion program. He holds an MBA from the University of Illinois at Chicago.

Chad Vincent is a lean manufacturing specialist and corporate operational excellence team member with Greif, Inc. in St. Louis, Missouri, where he provides professional guidance, facilitates implementation, and coaches and trains personnel at manufacturing facilities on lean enterprise and operational excellence. Prior to Greif, he worked as a quality engineer in the medical device field and as a project engineer in the construction industry; he also has worked in management in the logistics and transportation industry.

Chad has served as a senior lead examiner and judge for the Governor's Quality Award (Arkansas' equivalent of the Malcolm Baldrige National Quality Award) and is past president of the Leadership Arkansas Alumni Association Board of Directors. Chad is an advocate for ASQ's Socially Responsible Organization (SRO) Initiative and a member of SME's Lean to Green Committee. He has written articles, such as "Back in Circulation" for *Quality Progress*, on the utilizing of lean to achieve socially responsible and environmentally favorable results within organizations and on defining the 8 Green Wastes for environment, health, and safety (EHS) professionals to apply lean in their areas of expertise. He is a voting member of the US Technical Advisory Group (TAG) for the ISO 26000 Guidance for Social Responsibility.

Chad earned a BS in engineering management, specializing in quality engineering, from the Missouri University of Science and Technology, in Rolla, Missouri. He is a senior member of ASQ and SME, and is currently the ASQ Lean Enterprise treasurer and serves on the SME/AME/Shingo/ASQ Lean Certification Oversight and Appeals Committee. Chad is a CQE, CRE, CMQ/OE, and CSSBB, and he is Lean Bronze Certified.

Module 1

Cultural Enablers

- 1.1. Principles of Cultural Enablers
 - 1.2. Processes for Cultural Enablers
 - 1.3. Cultural Enabler Techniques and Practices
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Culture is the widening of the mind and the spirit.

—Jawaharlal Nehru

The first section of the Lean Body of Knowledge is dedicated to culture. Although lean is about the tools, it is more about creating a culture of people who truly believe in continuous improvement. This portion of the book focuses on what it takes to create, change, and lead an organizational culture into operational excellence. There is no lean without people. This section explores the importance of leading with humility, showing respect for people, having a well-crafted plan with a sense of urgency, and developing the people and leaders in your organization.

1.1

Principles of Cultural Enablers

According to Pascal Dennis (2007, 145), author of *Lean Production Simplified*, “Intensity is the soul of lean production, and team members are its heart.” In other words, people are the most critical element of lean production, and the *culture* the team members create is the major source of fuel required to propel lean systems forward in any organization.

In a vast majority of cases, the success or failure of any lean, Six Sigma, or other corporate initiative will depend on the people who execute it rather than on any equipment, consultant, software, or other tools and techniques. Thus, organizations that consider people as the prime appreciating asset and invest adequate time, effort, and money in hiring and developing the right people will get unmatched results (Dalal 2011, 584).

What Is a Culture?

A *culture* is the sum total of all behaviors, relationships, comprehension, and interactions that fuel overall alignment via collective thoughts, words, and actions.

What Is a Lean Culture?

Lean is an approach to improve quality, increase productivity, reduce costs, and increase customer satisfaction by eliminating waste and creating value.

A *lean culture* is the sum total of all the lean tools, techniques, and knowledge that exist within an organization at the root level and that fuel the overall organizational alignment via collective lean thoughts, words, and actions toward the elimination of waste and the creation of value.

Organizations that have a strong lean culture do two things:

1. They promote at least five key *cultural enablers* (safety, standards, leadership, empowerment, and collaboration), which allows the lean culture to exist
2. They build their business on the core fundamentals of respect for individuals

An Example of a Strong Lean Culture

The consistent growth, prosperity, innovation, and operational excellence of Toyota are clearly results of the Toyota Production System (TPS), which is built